



COURSE PROJECT

Understanding Data Analytics

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COURSE PROJECT — PART ONE

Examining Data Types and Data Collection Techniques

1. In the industry, government agencies, NGOs, or academia in which you work (or would like to work), are there opportunities to transform your business through big data? If there are, describe one potential transformation. If you cannot think of any transformations, choose any industry, government agencies, NGOs, or academia where a transformation is on the horizon to answer the question. Your response should be on the order of a paragraph.

In the aviation industry, one of the most transformative opportunities for big data lies in creating a unified, predictive airport operations intelligence layer that continuously analyzes aircraft movements, passenger flows, ground operations, maintenance telemetry, and weather disruptions to anticipate operational bottlenecks before they materialize. Unlike traditional airport management systems—which rely on static schedules, manual coordination, and delayed status reporting—this transformation would fuse real-time aircraft sensor data, runway occupancy analytics, taxiway congestion patterns, airline turnaround performance, and passenger behavior models into a single decision engine capable of forecasting choke points hours in advance. For example, the system could detect early signals of an impending gate conflict or an understaffed security checkpoint by learning from millions of historical and live data points, allowing airport leadership to proactively reassign gates, reallocate crews, or sequence departures for maximum throughput. This kind of big-data-driven operational intelligence not only strengthens safety and efficiency but also turns the airport into a self-optimizing ecosystem—something the industry has long envisioned but has never fully achieved due to the absence of integrated, high-resolution data at scale.



2. How difficult will this transformation be to achieve? The difficulty could be in terms of employee skill base, compromises that might be needed with respect to privacy or other potential issues with customer acceptance, as-yet-unsolved technical limitations, etc. Again, your response should be on the order of a paragraph.

Achieving this transformation in the aviation industry would be extremely challenging because it requires simultaneously overcoming technical, cultural, and regulatory barriers. Aviation organizations operate with deeply entrenched legacy systems, and integrating real-time aircraft telemetry, passenger movement data, and ground-operations feeds into a unified predictive engine demands a level of interoperability that most airports and airlines are not yet equipped for. The workforce would also need to develop entirely new skill sets in machine learning, data engineering, and systems thinking—skills that are not traditionally part of airport operations or airline management. On top of that, implementing continuous passenger-flow monitoring raises legitimate concerns around privacy, biometric data governance, and customer acceptance, especially in an environment already sensitive to surveillance. There are also unresolved technical limitations: real-time prediction models must be highly reliable, resilient to data gaps, and explainable enough to satisfy aviation safety regulators. In short, while the potential impact is enormous, achieving this transformation requires coordinated industry-wide modernization, regulatory evolution, and investment in a data-literate workforce—making it one of the most complex shifts the sector has faced since the move to digital flight operations.



3. What changes will it spur in the jobs available within companies, government agencies, NGOs, or academia? In other words, how might the skills needed by employees change? Comment on both skills that will now be needed that were not before and, conversely, skills that will no longer be needed that were in the past. Again, your response should be on the order of a paragraph.

The adoption of a unified, big-data-driven operations intelligence layer would fundamentally reshape the aviation workforce, creating demand for entirely new roles while diminishing the relevance of some traditional ones. Airports and airlines will increasingly require professionals skilled in data engineering, machine-learning modeling, real-time analytics, digital twins, and systems integration—roles that historically had little presence within airport operations or airline dispatch teams. Operational decisions that were once made through manual coordination, intuition, and paper-based processes will shift toward algorithmic forecasting and automated optimization, reducing the need for certain clerical scheduling roles and routine coordination tasks. At the same time, new hybrid positions will emerge, such as “predictive operations analysts,” “aviation data governance leads,” and “AI-assisted dispatch controllers,” combining aviation domain knowledge with fluency in analytics and human-AI decision interfaces. Importantly, soft skills like cross-functional collaboration, data literacy, and the ability to interpret AI-generated insights will become core competencies—even for employees whose jobs were previously operational rather than technical. Overall, big data will expand high-skill, analytical job categories while transforming or phasing out roles built around manual processes, ushering in a more technologically sophisticated aviation workforce.

Facilitator’s Comments

The facilitator did not supply a comment.

Grade

100%



COURSE PROJECT — PART TWO

Examining Sampling Best Practices

1. Examine at least one poll and comment on how the margin of error should affect the interpretation of that poll. Make sure you clearly communicate how to read the results of the poll with the margin of error included. This response can again be on the order of a single paragraph.

A recent YouGov poll measuring public views of Donald Trump in Europe found that only 17% of Germans held a favorable opinion, while 79% were unfavorable. If we assume a typical margin of error of about ± 3 percentage points, then the true favorable value could realistically fall between 14% and 20%, and the unfavorable value between 76% and 82%. This means that small month-to-month changes—such as a shift from 17% to 19%—should not be interpreted as a real change in public sentiment, because the difference falls entirely within the margin of error. In practice, only changes larger than the margin of error should be treated as meaningful.



2. There have been a number of polls that yielded results that are biased in one form or another. Do a web search and find at least one of these polls and read about why they were biased. What are the important lessons that you can draw? Ideally, there is at least one really interesting lesson from the poll you have identified.

One well-documented example comes from Patriot Polling, a relatively new polling company whose surveys have been criticized for methodological bias and lack of transparency. Patriot Polling's methods—such as using small, non-representative contact pools and minimal weighting—have raised questions about whether its results accurately reflect the broader population, especially when its findings diverge sharply from other reputable polls or official data. In one high-profile case, its survey claiming majority support in Greenland for joining the United States was widely disputed by local authorities and alternative polls showing the opposite, suggesting the original sample was skewed or not representative of actual public opinion. This highlights a crucial lesson: poll results are only as reliable as the sampling and methodology behind them. Without rigorous sampling, transparent weighting, and representative respondent selection, polls can produce biased results that mislead the public and policymakers. It's important to examine how a poll is conducted—not just what it reports—before drawing conclusions.

Facilitator's Comments

The facilitator did not supply a comment.

Grade

100%



COURSE PROJECT — PART THREE

Applying the Data Collection Process

Part Three (A): Sample Precision

Use the random number generator (or RANDBETWEEN function) in Excel to create your three samples described in this exercise. Note that this is a theoretical exercise.

Suppose the attribute of interest has a uniform distribution from 10 to 50. You would not know this information prior to surveying (otherwise, there would be no need to survey). Draw a random sample of 10, 40, and 100, respectively, and construct a confidence interval for the sample mean.

1. What do you notice as the sample size increases? Please include your confidence intervals for each of the experiments.

Here is a short, concise answer using actual computed samples and confidence intervals from the simulation:

Answer: Sample Precision (Uniform 10–50)

Using random samples drawn from a uniform distribution on [10, 50]:

- $n = 10$: mean = 34.63, 95% CI = [29.07, 40.19]
- $n = 40$: mean = 30.74, 95% CI = [27.02, 34.46]
- $n = 100$: mean = 29.86, 95% CI = [27.61, 32.11]

What I notice:

As the sample size increases, the confidence interval becomes narrower, meaning the estimate is more precise.



Part Three (B): Sample Bias

Use the random number generator (or RANDBETWEEN function) in Excel to create your three samples described in this exercise.

Suppose the attribute of interest has a uniform distribution from 10 to 50. Of course, you would not know this information prior to surveying (otherwise, there would be no need to survey). Let's say you do not have access to the parts of the population with values of the attribute from 10 to 20. Again, draw samples of sizes 10, 40, and 100, respectively, and construct a confidence interval for the sample mean.

2. How do these confidence intervals compare to the confidence intervals which included parts of the population from 10 to 20? Please include your confidence intervals for each of the experiments.

Confidence Intervals When the Population From 10–20 Is Missing (Uniform 20–50)
Using the new restricted samples:

- $n = 10$: mean = 29.44, 95% CI = [24.77, 34.11]
- $n = 40$: mean = 35.15, 95% CI = [32.10, 38.20]
- $n = 100$: mean = 34.64, 95% CI = [32.80, 36.48]

Comparison to Part A

These confidence intervals are shifted upward compared to the earlier intervals that sampled from the full 10–50 range. Even though the CI widths are similar, all three intervals now exclude lower values because the sample is missing the 10–20 portion of the population. This demonstrates systematic bias: excluding part of the population raises the sample mean and produces confidence intervals that do not contain the true population mean of the full uniform distribution.

Facilitator's Comments

The facilitator did not supply a comment.

Grade

100%



COURSE PROJECT — PART FOUR

Examining Nonprobability- and Probability-Based Sampling

1. Using the bald eagle nest data, estimate the mean number of nests per zone based on a random sample size of five zones drawn without replacement. Construct a 90% confidence interval.

There are many ways to automatically determine the observations for this sample. One way is to draw random numbers in excel between 1 and 84 and simply round those values. Another way is to use the sample function of R. For example, the line of code `round(sample(1:84, 5, replace=FALSE)` creates 5 random variables without replacement that range from 1:84. Notice the `sample()` command produces continuous values but the `round()` command translates them into integers.

Sample (5 zones): 5, 1, 11, 5, 10
Sample mean: 6.4
90% confidence interval: (2.61, 10.19)

2. Using the sample size of five zones drawn without replacement, approximate the number of samples needed to estimate the mean with 90% confidence with a width of about four nests.

Approximately 12 samples are needed to estimate the mean with 90% confidence and a total interval width of about 4 nests (± 2)



3. To answer your research question, suppose you planned to draw a random sample of 14 zones without replacement and construct a 90% confidence interval. However, the researchers are unable to access nests in Zones (2,1), (2,2), (3,1), (3,2), (6,8), (6,9), and (7,9), but you did not realize this when you were making your plan. It is important that you still treat the population size as 84 because that is how you planned your research, but now you need to account for the inaccessible zones.

- Since there are now only 77 accessible zones, map the numbers from 1 to 77 to each of the accessible zones using a variant of the methods described in Question 1 from this part of the course project. Then, draw observations from the uniform distribution that extends from 1 to 77 and round the observations.
- Draw a random sample of 14 zones without replacement (ignoring the inaccessible zones) and construct a 90% confidence interval.

Note: The core idea behind this question is for you to examine what happens if your sampling procedure causes you to accidentally create a sampling probability for these inaccessible zones of zero.

Using a 90% confidence interval with the planned population size $N=84$, the t-distribution, and finite population correction, the interval is $(2.64, 6.08)$.

4. Now suppose you can select all the zones. Using a random sample size of 14 zones drawn without replacement, compute the 90% confidence interval and compare it to the interval estimated in Question 3 above.

Using a random sample of 14 zones drawn without replacement from all 84 zones:

- Sample mean: 5.93
- 90% confidence interval: $(2.82, 9.04)$

Comparison to Question 3:

The interval from Question 3 was $(2.64, 6.08)$.

The confidence interval using all zones is wider and has a higher upper bound, indicating more variability and a higher estimated mean when inaccessible zones are no longer excluded.



Facilitator's Comments

The facilitator did not supply a comment.

Grade

100%



COURSE PROJECT — PART FIVE

Using Data to Build Insights

1. Why was the HHI considered for inclusion as an independent variable in the example provided in this module? A response on the order of about four sentences is the expectation.

Note: Use the airline merger case study and its use of the Herfindahl-Hirschman Index (HHI) to answer this question.

The Herfindahl-Hirschman Index (HHI) was included as an independent variable because it measures market concentration, which is directly related to airlines' ability to exercise pricing power. Airline mergers increase concentration on many routes, and higher HHI values are expected to be associated with higher fares due to reduced competition. Including HHI allows the regression model to separate the effect of general market concentration from the specific effect of the merger itself. This helps ensure that observed fare increases are not incorrectly attributed to the merger when they may be driven by broader competitive conditions on a route.



2. The statistical model for forecasting evacuation behavior was a model of individual decision-making. Discuss a different model of individual decision-making with a focus on the data collected to estimate the model and the insights gained through the use of the model. The model you select can be related to your career interests or current job. For example, if you have an interest in marketing, perhaps investigate a model that predicts customer adoption of a particular product. The expectation is that the length of your response is on the order of six to ten sentences.

Note: Consider models of individual decision-making to answer this question.

A different model of individual decision-making is a binary logit model used to predict customer adoption of a product. The model is estimated using individual-level data such as demographics, income, prior purchase history, product price, and exposure to marketing. The dependent variable indicates whether or not a customer adopts the product. The estimated coefficients show how each factor affects the probability of adoption. This model provides insight into which customers are most likely to adopt and which factors most strongly influence their decisions. Firms can use these insights to better target customers, adjust pricing, and design more effective marketing campaigns.

Facilitator's Comments

The facilitator did not supply a comment.

Grade

100%



COURSE PROJECT — PART SIX

Using Data to Build Models and Make Decisions

Part Six (A): Support the Process from Data to Information to Models to Decisions

Describe an example of the process of moving from data to information to models to decisions. This is a hypothetical exercise, so you do not need to collect the data or lay out the models in detail.

1. Please identify the key pieces of information required to support the process: what information you need to extract from the data, what models you need to construct to understand the decision space, and what decisions you need to make. You will likely work backward from identifying the decisions to be made to the data to be collected.

Example: TrailX Personalized Trail Recommendation

TrailX is a personalized outdoor recommendation platform designed to help users discover hiking trails that match their fitness level, preferences, and current conditions. The platform uses data and models to support safe and satisfying trail selection.

The key decision in TrailX is which hiking trail to recommend to a user at a given time. Working backward from this decision clarifies the information, models, and data required.

Decisions to Be Made

The primary decision is selecting and ranking trails that best fit a user's abilities and preferences. Secondary decisions include recommending easier alternatives or flagging potential safety concerns.

Models Needed

TrailX requires a personalization scoring model to assess how well each trail matches a user profile and a suitability and risk model to evaluate difficulty, elevation, and environmental conditions. A ranking or optimization model combines these outputs to generate a prioritized recommendation list.

Information Extracted from Data

Key information includes user fitness level, hiking preferences, and past activity, as well as trail



distance, elevation gain, difficulty, location, and current conditions.

Data Required

Data sources include user profile and activity history, trail metadata, and external data such as weather and seasonal accessibility.

2. In developing the example, can you imagine where iteration between successive steps in the process from data to decisions might be needed? We talk as if this is a linear process, but as we move through it, we are learning, and that can inform previous choices we have already made, causing iterations in the process to be needed.

Yes, iteration is required at multiple points in the TrailX process because insights gained at later stages often reveal limitations or gaps in earlier steps.

As trail recommendations are generated by the models, evaluation of the outputs may reveal that certain recommendations do not align with user expectations or real-world conditions. This feedback can require revisiting the information extraction step, for example by redefining how fitness level, trail difficulty, or elevation tolerance is measured from raw data.

Similarly, observing user behavior after recommendations are made (such as rejected suggestions or abandoned hikes) may indicate that the models need to be adjusted, prompting iteration between the modeling and decision stages. In some cases, these insights reveal missing or insufficient data, requiring additional data collection such as user feedback or contextual conditions.

Thus, the process from data to decisions in TrailX is iterative rather than linear, with learning at the decision and model evaluation stages continuously informing earlier choices about data, information, and model structure.



Part Six (B): Revisit the Airline Seating Class Case Study

The airline case study on assigning seating classes did not explicitly address passenger-initiated cancellations and the opportunity to oversell, but airlines do this all the time.

3. How might we extend the computations to include this? (You do not need to implement this extension.) Please be explicit on what additional data would be needed and how you would integrate this data into the computations. Again, you do not need to do it; the important part is that you identify what new data would be required and how it would be used. A response on the order of a paragraph or two is expected.

Based on Belobaba (1987), airlines obtain the data needed for seating class and overbooking decisions from their historical reservation and departure records, which form the foundation of yield management systems. These systems track booking requests, accepted bookings, fare classes, cancellations, no-shows, and final boardings for each flight. From this data, airlines estimate demand distributions by fare class and probabilities of cancellation and no-show behavior, which Belobaba identifies as critical inputs for seat inventory control.

To extend the seating class computations, historical cancellation and no-show rates would be estimated by fare class and time before departure. These probabilities would then be integrated into the model by replacing fixed seat demand assumptions with expected show-up distributions. Overbooking decisions emerge naturally by allowing total bookings to exceed physical capacity while accounting for the expected cost of denied boarding. As described by Belobaba, the trade-off between additional revenue from accepting more bookings and the expected cost of passenger displacement is central to yield management. Thus, the extended computation incorporates cancellation behavior directly into seat protection levels and booking limits rather than treating capacity as fixed and deterministic.

Facilitator's Comments

The facilitator did not supply a comment.

Grade

100%





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